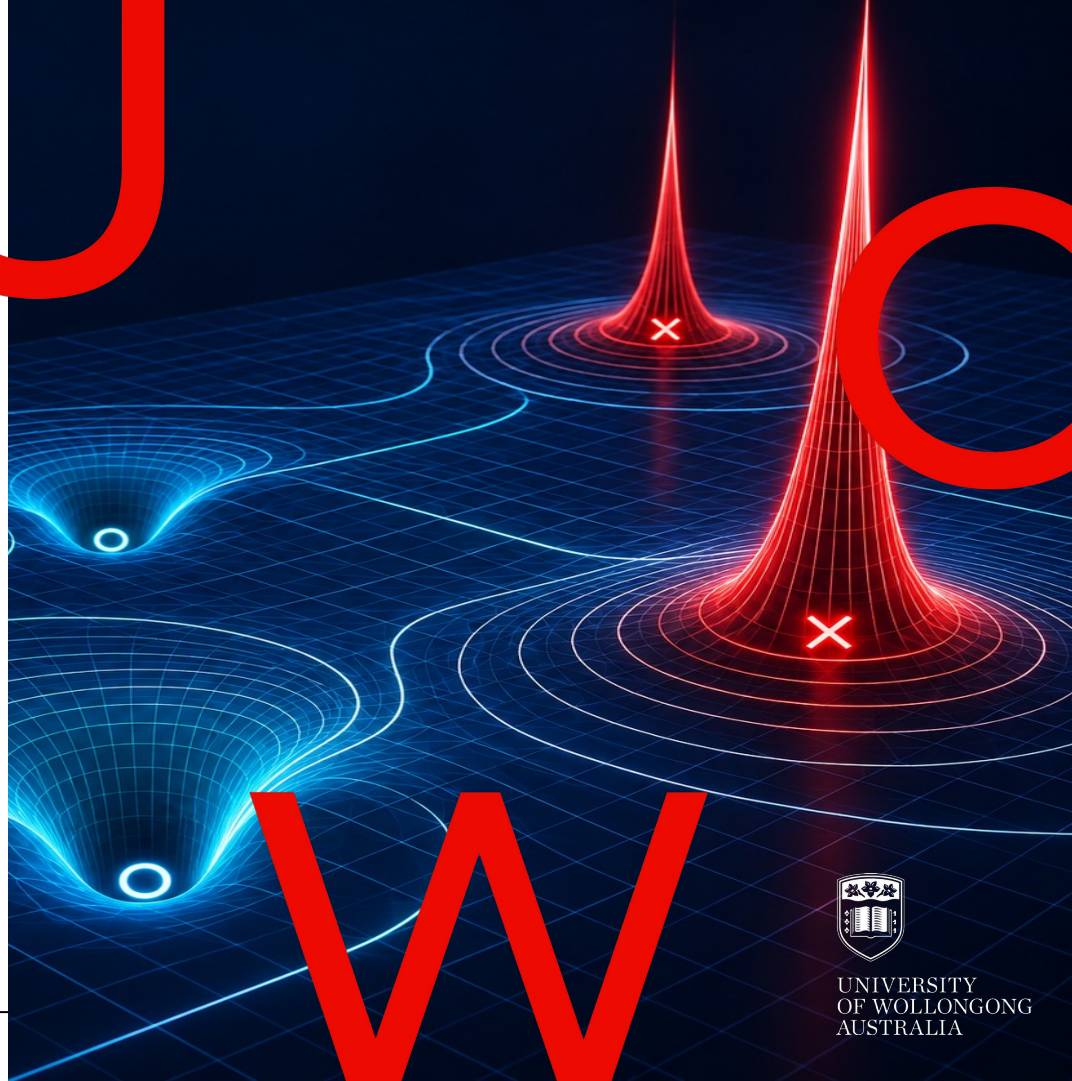
MECH321
Dynamics of Engineering Systems

Lecture 8
Nyquist Stability and Margins

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PART 1

Nyquist curves and stability

01 Critical point

02 Encirclement convention

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Stability margins

Model uncertainty

Parameters drift with temperature, load and age

Some dynamics were never modelled at all

Gain tolerance

A stable loop may tolerate only a 20% gain increase.

Additional dynamics

Sensor filters and sensing delay

Additional phase lag

What a margin is

Gain or phase change required to reach a stability boundary.

Why the critical point is -1

CORE

Closed-loop poles satisfy $1 + L(s) = 0$ **that is** $L(s) = -1$

In terms of the frequency response, $L(j\omega) = -1$ means magnitude exactly 1 and phase exactly $-180°$ at the same frequency.

Physically

At the critical frequency, the returned signal reinforces the loop, allowing sustained oscillation.

Phase crossover: $-180°$. GM and PM definitions: slides 12–13.

Critical point for $L(j\omega)$: -1.

The Nyquist contour

OPTIONAL

The contour in the s-plane

Up the **whole imaginary axis**, from $-j\infty$ to $+j\infty$

Closed by a very large semicircle through the right half-plane

With small detours around any pole sitting on the imaginary axis

Why the detours matter

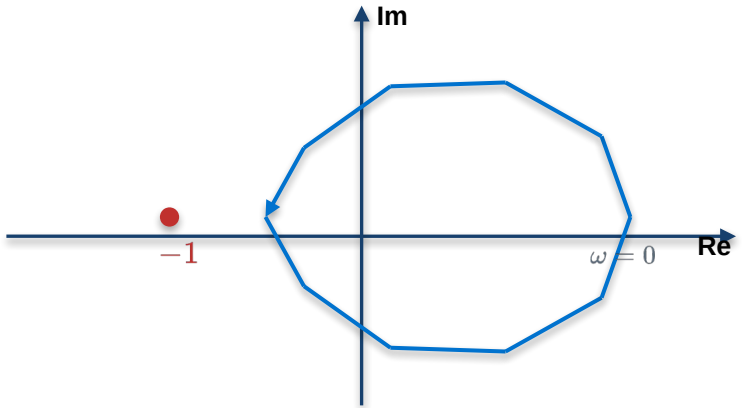
A loop with an integrator has a pole at the origin

The detour maps to a large arc in the L-plane

Miss it and the encirclement count comes out wrong

Include negative frequencies, their conjugate branch and the closing contour.

The Nyquist plot



What is plotted

$L(j\omega)$ as a single curve in the complex plane, as ω runs from $-\infty$ to $+\infty$

Magnitude: distance from origin. Phase: angle.

How many times does the curve wrap around the point -1, and in which direction?

Use the full Nyquist count for unstable open-loop plants or multiple crossovers; classical margins alone may be insufficient.

The counting relation

CORE

$$Z = N + P$$

P

Open-loop poles in the right half-plane

Open-loop model: $L(s)$

N

Encirclements of -1 by the Nyquist locus

Counted clockwise, under the convention used here

Z

Closed-loop poles in the right half-plane

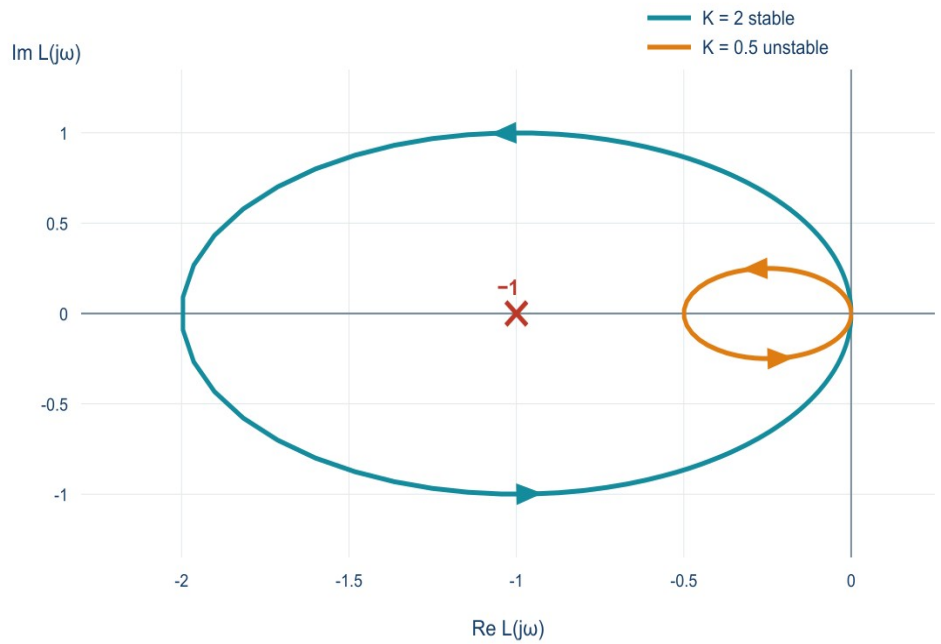
$Z = 0$ is stability

Anticlockwise-positive texts use $Z = P - N$. State the convention.

Nyquist count for an unstable plant

EXAMPLE

$K/(s-1)$: one right-half-plane open-loop pole $P = 1$



$P = 1$: open-loop pole at +1.
$K = 2$: anticlockwise loop, $N = -1$.
$Z = P + N = 0$: stable.
$K = 0.5$: **no encirclement**, $N = 0$.
$Z = 1$: unstable.

Cross-check: 1 + $L(s) = 0$ gives $s = 1$ − K.
The pole is −1 for $K = 2$; +0.5 for $K = 0.5$.

Locate a stability boundary without a plot

EXAMPLE

Form the closed-loop polynomial

For $L(s) = \frac{K}{s(s+1)(s+2)}$, unity feedback gives characteristic polynomial $s^3 + 3s^2 + 2s + K$. The open loop has no right-half-plane poles, so $P = 0$; it also has a pole at the origin, which requires a small Nyquist detour.

Test an imaginary-axis root

At the boundary: $s = j\omega$.
Real and imaginary parts: $K - 3\omega^2 = 0$ and $2\omega - \omega^3 = 0$.
Nonzero solution: $\omega = \sqrt{2}$ rad/s, $K = 6$.
Root check at $K = 1$: -2.325 and $-0.338 \pm j0.562$ (all LHP).
Stable range: $0 < K < 6$; above K=6, a pair enters the RHP. Nyquist reaches −1 at $K = 6$.

CHECK At a critical crossing the gain margin has collapsed to one (0 dB).

PART 2

Stability margins and design evidence

01 Gain and phase margins

02 Manual Nyquist check

03 Delay and release evidence

Gain margin

At the phase crossover ω_{pc}, where $\angle L(j\omega) = -180°$: $GM = \frac{1}{|L(j\omega_{pc})|}$ **or in decibels** $GM_{dB} = -|L|_{dB}$

What it means

Multiply the loop gain by GM and the curve reaches -1 exactly

Nyquist applicability: slide 7.

Typical target

6 to 12 dB, that is a factor of 2 to 4

Check the specified gain uncertainty.

When it is infinite

If the phase never reaches $-180°$, there is no crossover

Still check PM and Nyquist stability.

Phase margin

At the gain crossover ω_{gc}, where $|L(j\omega)| = 1$ (that is 0 dB): $PM = 180° + \angle L(j\omega_{gc})$

What it means

Add PM degrees of extra lag and the loop reaches -1

A delay T costs $\omega_{gc}T$ radians of phase

Typical target

40° to 60°

Verify damping from the actual closed-loop response.

60°: approximate $\zeta \approx 0.6$ (see slide 19).

$\zeta \approx \frac{PM}{100}$ (dominant-pair approximation; slide 19).

Reading margins off a Bode plot

CORE

1. **Find where the magnitude crosses 0 dB**
   That frequency is ω_{gc}
2. **Read the phase there**
   $PM = 180°$ plus that phase, which is negative
3. **Find where the phase crosses $-180°$**
   That frequency is ω_{pc}
4. **Read the magnitude there**
   GM in dB is minus that value

Multiple crossovers
Multiple 0 dB or $-180°$ crossings: use full Nyquist counting (slide 7).

Worked: reading stability margins

EXAMPLE

What was read off the plot

At $\omega_{gc} = 3$ rad/s: phase $= -140°$

At $\omega_{pc} = 8$ rad/s: magnitude $= -10$ dB

The two margins

$PM = 180° - 140° = 40°$

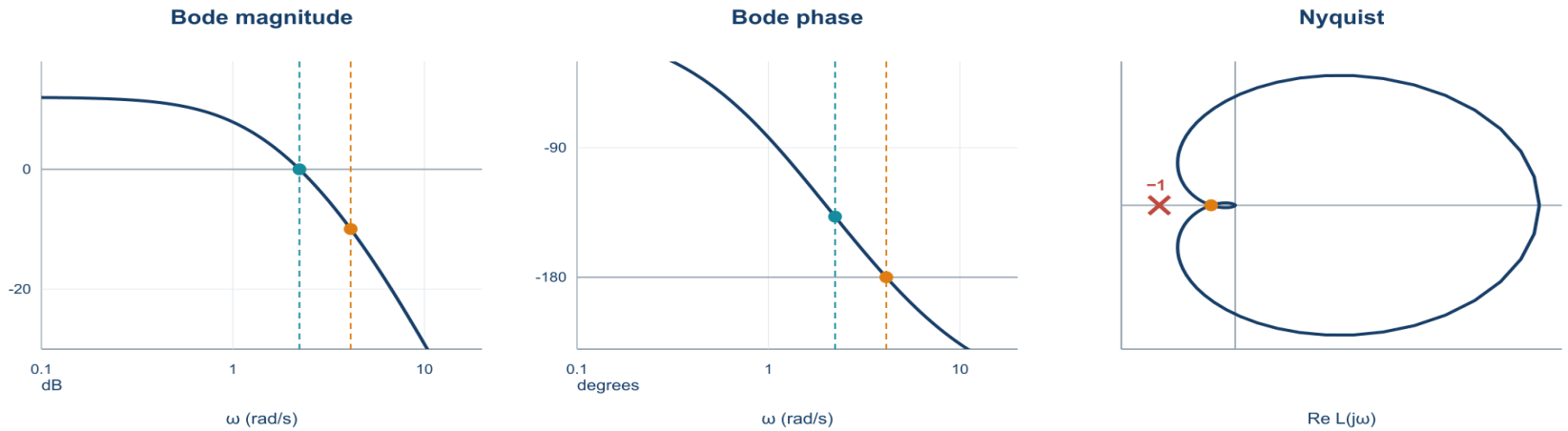
$GM = 10 \text{ dB} = 10^{\left(\frac{10}{20}\right)} = 3.162$

Report crossover frequencies

Equal PM at 3 and 300 rad/s gives different delay tolerance.

One loop: Bode, Nyquist and closed-loop poles

EXAMPLE



$L(s) = 40/[(s+1)(s+2)(s+5)], P = 0$
$\omega_{gc} \approx 2.23$ rad/s; phase ≈ −137.9°; $PM \approx 42.1°$
$\omega_{pc} = \sqrt{17} \approx 4.12$ rad/s; $|L| = 40/126$; $GM = 3.15 \approx 10.0$ dB

Nyquist: $N = 0$, $Z = 0$. Closed-loop roots: −6.57 and −0.715 ± 2.66j.

Manual Nyquist Construction

EXAMPLE

Model from slide 16

$$L(s) = \frac{40}{(s+1)(s+2)(s+5)} = \frac{40}{s^3+8s^2+17s+10}$$

$$A = 10 - 8\omega^2, \quad B = 17\omega - \omega^3$$

$$L(j\omega) = \frac{40(A-jB)}{A^2+B^2}$$

$$\operatorname{Re} L = \frac{40A}{A^2+B^2}, \quad \operatorname{Im} L = \frac{-40B}{A^2+B^2}$$

$$\omega_{pc} = \sqrt{17}, \quad L\left(j\sqrt{17}\right) = -\frac{40}{126} \approx -0.3175$$

$$L(0) = 4, \quad \lim_{\omega\to\infty} L(j\omega) = 0$$

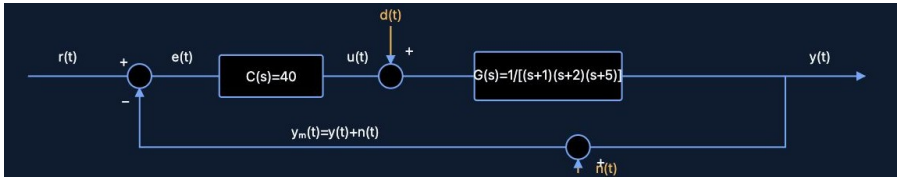
$$L(-j\omega) = \overline{L(j\omega)}, \quad P = 0,\ N = 0,\ Z = P + N = 0$$

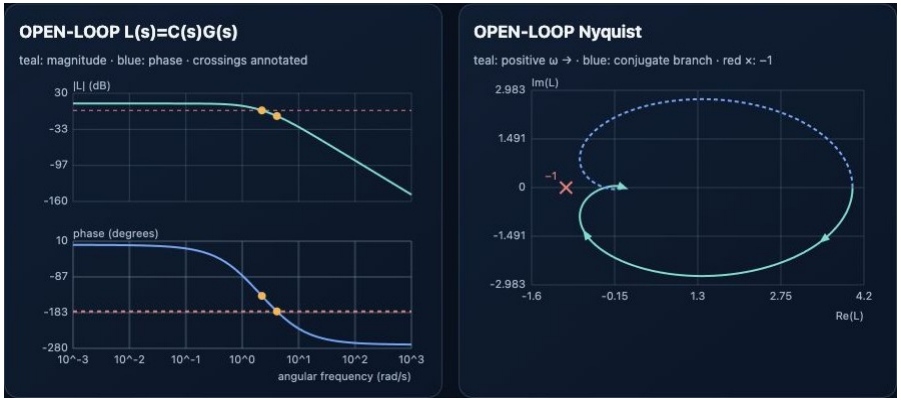
Poles −1, −2, −5: P=0. No imaginary-axis detour.
Positive frequency: 4 ⟶ lower half-plane ⟶ −0.3175 ⟶ upper half-plane ⟶ 0.

Reflect in the real axis for negative frequencies.
Mark both traversal directions.
No encirclement of −1: the closed loop is stable.

Live demo: gain, margins and stability

OPEN LIVE DEMO ↗





Feedback Explorer: Margins

Slide 16 loop: $G(s) = \frac{1}{(s+1)(s+2)(s+5)}$, $K_c = 40$.

1 Read GM and PM. Try $K_c = 80$, 126, then 160.

2 At each gain, compare −1 on Nyquist, closed-loop poles and the step response.

MATLAB alternative (Control System Toolbox)

```
s=tf('s'); K=40;
G=1/((s+1)*(s+2)*(s+5)); L=K*G;
T=feedback(L,1);
figure; margin(L); grid on
figure; nyquist(L); grid on
figure; step(T,0:.01:12); grid on
pole(T)
% Interactive: controlSystemDesigner(G,tf(K))
```

CHECK $K = 40$: $GM \approx 10.0$ dB, $PM \approx 42.1°$. $K = 126$: $L\left(j\sqrt{17}\right) = -1$ and poles -8, $\pm j\sqrt{17}$. $K = 160$: two RHP poles, so the loop is unstable.

Phase margin and damping

CORE

Phase margin	Approximate ζ	Overshoot
20°	0.2	53 %
40°	0.4	25 %
60°	0.6	9 %
70°	0.7	5 %

Approximation conditions

$\zeta \approx \frac{PM}{100}$ is a dominant-pair, single-crossover guide. Verify the actual step response.

What a margin does not guarantee

Multiple crossovers

A resonant plant can cross 0 dB three times

Check every crossover.

Closest approach to −1

The Nyquist curve can pass close to -1 while both classical margins look adequate

Conditional stability

Some loops become unstable if the gain is reduced

A gain margin quoted upward says nothing about that

Vector margin

Vector margin: closest Nyquist distance to -1. Inspect alongside GM and PM.

EXAMPLE

Uncertain resonance: use a worst case

A nominal margin can disappear near a lightly damped mode

Measured scenario

A soft gripper has nominal 0 dB crossover at 12 rad/s with phase −135°, so **nominal PM is 45**°. A credible low-damping test still crosses near 12 rad/s but has phase −170°: **PM falls to 10**°. The low-frequency tracking curve barely changes, hiding the hazard.

Engineering response

Do not raise gain from the nominal result. Record the resonant frequency and damping range, measure or model phase around the mode, and recompute margins at each credible extreme. If the minimum margin is unacceptable, reduce crossover, add damping or change the controller, then check effort and response again.

Delay and phase margin

phase lost $= \omega_{gc} \cdot T$ **radians** $= 57.3 \cdot \omega_{gc} \cdot T$ **degrees**

a pure delay T costs phase **in proportion to the crossover frequency**

ω_{gc}	$T = 5$ ms	$T = 20$ ms	$T = 50$ ms
3 rad/s	0.9°	3.4°	8.6°
30 rad/s	8.6°	34°	86°
100 rad/s	29°	115°	286.5°

Same delay: crossover 3 versus 100 rad/s gives proportionally different phase loss.

A delay budget from phase margin

EXAMPLE

Given delay: phase loss

A pure delay contributes phase lag of ωT radians at frequency ω. For a separate example, assume $\omega_{gc} = 4$ rad/s and $PM = 40°$. A 0.10 s sensing/computation delay adds 0.4 rad, or 22.9°, of lag at crossover. The remaining approximate margin is only 17.1°.

Delay boundary and reserve

The mathematical delay that consumes all 40° is $(40\pi/180)/4 \approx 0.175$ s, provided crossover stays near 4 rad/s. **Do not design to that boundary**: uncertainty, extra poles and multiple crossovers need reserve. Recompute the full delayed loop response before release.

CHECK **The same 0.10 s delay costs twice the phase if crossover doubles to 8 rad/s.**

MATLAB Stability Checks

EXAMPLE

PREDICT

Where does the curve start, at $\omega \to 0$? Magnitude and phase

Where does it end, at $\omega \to \infty$?

Track the phase and traversal direction.

Roughly where will it cross the negative real axis?

COMPUTE

Then check

```
[GM, PM, Wpc, Wgc] = margin(L)
margin(L)   % draws the Bode plot with both marked
nyquist(L)  % and look at the approach to −1
```

Compare GM, PM and both crossover frequencies with the sketch.

Use allmargin(L) to list all crossover pairs.

Margins as engineering evidence

CORE

Report

GM in dB and as a factor

PM in degrees

Both crossover frequencies

The model the margins were computed from

Also report

Margins at the worst credible parameters, not only the nominal ones

The frequency range over which the model was validated

Then verify

The step response and its overshoot

Behaviour with the actuator limit in place

Sensitivity to the uncertain parameters

GM and PM describe specified gain/phase changes. Also verify response, actuator limits and uncertainty.


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